

AMAN RAJ

Bengaluru
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Profile Summary

Software Engineer with experience in C++, Modern C++ and Embedded Software development in Automotive domain. Skilled in Adaptive AUTOSAR, Linux, Software Integration, Integration Testing, Debugging and GitLab CI/CD workflows. Experienced in developing and validating scalable software systems, ECU communication and event-driven data processing architectures.

Work Experience

KPIT Technologies

Bengaluru, India

Software Engineer

July 2024 – Present

- Developed and maintained software modules using Modern C++ (C++11/14/17) for Automotive Embedded Systems.
- Contributed to Adaptive AUTOSAR based software integration and validation activities.
- Performed debugging, issue analysis and software verification in Linux environments.
- Executed integration testing and validation across embedded software components.
- Automated engineering workflows using Python and supported software quality improvements.
- Collaborated with cross-functional teams in Agile software development lifecycle.
- Supported GitLab CI/CD workflows and release activities.

Software Engineer Intern

Dec 2023 – July 2024

- Resolved MISRA C++ and AUTOSAR guideline violations using static analysis tools.
- Automated validation and verification workflows using Python.
- Completed training in Modern C++, AUTOSAR, Linux and Embedded Systems.

Education

B.Tech – Computer Science Engineering

Vellore Institute of Technology

CGPA: 7.82/10

Aug 2020 – May 2024

Class XII – BSEB

P.C.P Inter College

Percentage: 87.2%

Jul 2017 – Jul 2019

Skills

Programming Languages:

C++, Modern C++, Python, Java, SQL

Automotive & Embedded:

Adaptive AUTOSAR, ECU Integration, CAN, Ethernet, DAQ, Embedded Systems

Software Development:

Object Oriented Programming, Software Integration, Integration Testing, Debugging, CI/CD

Tools & Platforms:

Linux, Git, GitLab CI/CD, GDB, Vector CANoe, CAPL

Certifications

AWS Cloud Practitioner Essentials

Cybersecurity Essentials – Cisco

SQL Intermediate – HackerRank

Projects

Automotive Data Acquisition (DAQ) Platform

Current

- Worked on centralized vehicle data acquisition architecture for event-driven data collection.
- Implemented configuration-driven execution using JSON-based DAQ Specification Format (DSF).
- Processed event configurations and coordinated service activation based on DAQ event settings.
- Integrated communication across multiple ECUs including CAN, ADAS and OBC.
- Supported telemetry data acquisition and transmission to centralized backend communication systems.
- Contributed to software integration, debugging and validation activities.

Intravenous Drip Monitoring System – IoT

Mar 2023 – Aug 2023

- Developed a real-time IV monitoring system using NodeMCU and sensor integration.
- Built embedded data acquisition and dashboard monitoring workflow.

Leadership

IEEE MTTTS – VIT

May 2022 – Jan 2023

Technical Team Lead

- Led a team of 15+ members across technical workshops and engineering activities.
- Mentored junior members and improved collaboration and execution.

Matrix – VIT

May 2021 – Feb 2022

Technical Team Lead

- Managed project coordination and technical event execution.
- Improved planning, delegation and team productivity.

Achievements

All India Rank 260 in VITEEE

3rd position in district in Class XII examinations